

Université Catholique d'Afrique Centrale (UCAC)
Institut Catholique de Yaoundé (ICY)

COURSE TITLE: INFERENCE STATISTICS

DEPARTMENTS: BEM2

LECTURER: ARREY Marinus

INSTRUCTIONS: ANSWER ALL QUESTIONS

SECOND SEMESTER EXAMINATION

Question 1: (4 marks)

Explain the following concepts:

- a) Population parameter.
- b) Probabilistic and non-probabilistic sampling
- c) Type I and type II Error.
- d) Significance level (α)

Question 2: (6 marks)

Fifty Rabbits were given special diet for a certain period and control group of 50 other Rabbits were given normal diet. Their average gain in weight were found to be 8.4 kgs and 6.3 kgs respectively and the common S.D. for gain in weight was 2 Kgs. Assuming normality of the distributions would you conclude that the diet really promoted weight?

Question 3: (5 marks)

Given below are the gains in weights of Lions on two diets, X and Y

Diet X	15	20	24	24	18	15	18	12
Diet Y	16	24	12	20	30	25	28	

Test at 5% level, whether the two diets differ significantly with regard to increase in weight.

Question 4: (5marks)

Consider the information on the table below concerning the days of the week and the number of accidents on the Douala – Bafoussam road.

Days of the week	Sun	Mon	Tue	Wed	Thur	Fri	Sat
No. of accidents	16	24	12	20	30	25	28

Test whether the accidents occur uniformly over week days on the basis of the above information:

Good Luck!